UNIT 5 · 9 hours

78XX Series IC Voltage Regulators



Op-Amp and Linear IC Applications

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Explain the need for a regulated DC supply
- Describe the 78XX three-terminal fixed regulators
- Draw the standard connection with capacitors
- Compute regulator power dissipation

Syllabus context

Introduction, op-amp specifications, differential and common-mode operation, op-amp applications: inverting/non-inverting amplifier, summing, integrator, differentiator, comparator, 555 timer IC in astable mode, 78XX series IC voltage regulators.

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Why IC regulators?

- A rectifier + filter gives DC that varies with mains and load.
- A voltage regulator keeps VO constant against input, load and temperature variation.
- IC regulators are simple three-terminal devices with built-in protection.
- Better than a Zener regulator: higher current, lower output resistance.

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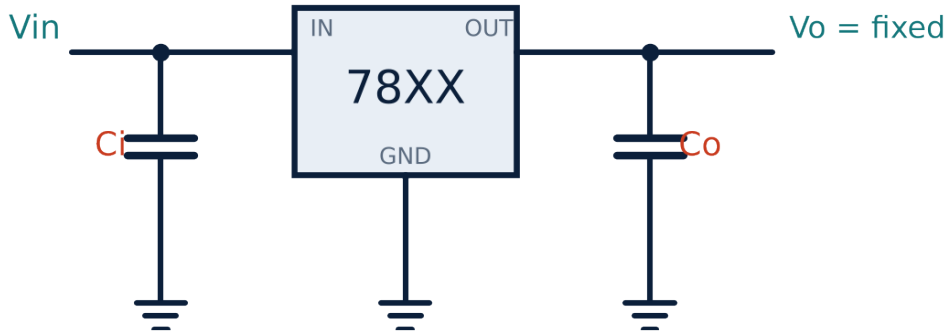
The 78XX family

Device	Output	Max current
7805	+5 V	1 A
7806	+6 V	1 A
7809	+9 V	1 A
7812	+12 V	1 A
7815	+15 V	1 A
79XX	negative voltages (7905 = −5 V)	1 A

The last two digits give the output voltage; the 78 prefix means positive fixed regulator.

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Standard connection



- Three terminals: IN, GND (common), OUT.
- Input capacitor Ci (≈ 0.33 µF): cancels lead inductance, stabilises.
- Output capacitor Co (≈ 0.1 µF): improves transient response.
- Vin must exceed Vout by the dropout (≈ 2 V): e.g. ≥ 7 V for the 7805.

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Features

Fixed output

Accuracy about ±4 %; line regulation ≈ 0.01 %/V.

Built-in protection

Thermal shutdown, short-circuit current limiting, safe-area protection.

Heat sinking

Power dissipated = (Vin − Vout) × IL; use a heat sink above ≈ 1 W.

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Numerical: dissipation of a 7805

Problem A 7805 is fed from 12 V and supplies 0.5 A. Find the power dissipated and the output power.

- Voltage across regulator = 12 − 5 = 7 V
- Pdiss = 7 V × 0.5 A = 3.5 W
- Pout = 5 V × 0.5 A = 2.5 W; efficiency ≈ 2.5/(12 × 0.5) = 42 %

Answer: Pdiss = 3.5 W: needs a heat sink

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Applications

- Power supplies for logic ICs (7805 for 5 V TTL/CMOS).
- On-board regulation in microcontroller boards.
- Adjustable output using a resistor from GND to ground (boost).
- Current source: with a fixed resistor, I = Vreg/R.

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Key takeaways

- 78XX: positive 3-terminal fixed regulators; 7805 = +5 V
- Need Vin ≥ Vout + ≈ 2 V
- Ci and Co improve stability
- Pdiss = (Vin − Vout) IL

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Practice questions

Q1. Draw the connection of a 7812 and explain the capacitors.

Q2. A 7809 with Vin = 15 V supplies 0.4 A. Find Pdiss.

Q3. What are the built-in protections of a 78XX regulator?

Q4. What does 7912 denote?